

Problem-Solution Fit

Date	02 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction Using Machine Learning
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] - Banks and Financial Institutions - Bank Credit Analysts - Loan/Credit Officers - Customers applying for credit cards	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] - Limited time for reviewing applications. - Heavy workload. - Dependence on manual verification. - Inconsistent decision-making.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Pros - Manual verification is simple to understand. - Existing banking systems are already in use. Cons - Time-consuming. - High operational cost. - Inconsistent decisions. - Difficult to scale. - Error-prone.
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2. PROBLEMS / PAINS
+ ITS FREQUENCY [PR]

- Manual verification is slow.
- Large number of daily applications.
- Human errors during approval.
- Difficulty identifying high-risk applicants.
- Frequent delays in approval decisions.

9. PROBLEM ROOT / CAUSE [RC]

- Manual verification of applicant information.
- Large volume of applications.
- Complex evaluation of income and credit history.
- Lack of automated prediction systems.

7. BEHAVIOR
+ ITS INTENSITY [BE]

- Analysts review hundreds of applications daily.
- Applicants expect quick approval decisions.
- Banks prioritize accuracy and low financial risk.

3. TRIGGERS TO ACT
[TR]

- Increasing number of credit card applications.
- Need for faster approval.
- Requirement to reduce financial risk.
- Demand for automated decision-making.

10. YOUR SOLUTION
[SL]
Credit Card Approval
Prediction Using Machine
Learning

- Automates the approval process.
- Preprocesses applicant and credit history data.
- Uses Logistic Regression, Random Forest, Decision Tree, and Gradient Boosting/XGBoost models.
- Predicts approval or rejection instantly.
- Deploys the best-performing model through a Flask web application.
- Reduces manual effort, improves consistency, and supports faster decision-making.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

- Flask Web Application
- Bank Internal Portal
- IBM Watson ML Deployment (if used)

OFFLINE

- Bank Branch
- Customer Service Desk

4. EMOTIONS
BEFORE / AFTER [EM]:
Before

Frustrated
Overloaded
Uncertain
Concerned about mistakes

After

Confident
Satisfied
Faster decision-making
Reduced workload